

Hospital Management Performance Report

Project Overview

This project analyzes hospital admissions data using SQL to evaluate patient volume, treatment costs, readmissions, length of stay (LOS), and resource utilization. The objective is to identify operational trends and provide actionable recommendations that support healthcare decision-making.

KPI Summary

Total Admissions

Result: 50 Admissions

Finding: The hospital recorded 50 admissions during the study period.

Insight: Admission volume reflects overall healthcare demand and resource utilization.

Recommendation: Monitor admission trends regularly to support staffing and operational planning.

Unique Patients

Result: 18 Unique Patients

Finding: A total of 18 unique patients generated 50 admissions.

Insight: Multiple admissions per patient suggest recurring healthcare utilization and potential readmission patterns.

Recommendation: Investigate drivers of repeat admissions and strengthen patient follow-up processes.

Top Diagnosis

Result: Essential Hypertension (9 Cases)

Finding: Essential Hypertension was the most frequently recorded diagnosis.

Insight: Hypertension represents a significant portion of the hospital's disease burden and may contribute to recurring healthcare utilization.

Recommendation: Expand hypertension management programs and patient education initiatives.

Top Department

Result: Cardiology (14 Admissions)

Finding: Cardiology handled the highest number of patient admissions.

Insight: Cardiology accounted for approximately 28% of all admissions, making it the busiest department.

Recommendation: Ensure adequate staffing and resource allocation within Cardiology to support patient demand.

Highest Cost Patient

Result: Imran Ali – 835,000

Finding: Imran Ali generated the highest cumulative treatment cost among all patients.

Insight: High-cost patients may require complex treatment plans, repeated admissions, or intensive resource utilization.

Recommendation: Review treatment pathways and identify major cost drivers among high-cost patients.

Average Treatment Cost

Result: 89,480

Finding: The average treatment cost per admission was 89,480.

Insight: Treatment costs represent a significant operational expense and directly affect hospital resource allocation.

Recommendation: Monitor high-cost diagnoses and evaluate opportunities for cost optimization.

Elderly Readmissions

Result: 6 Patients

Finding: Six patients aged above 60 experienced more than three admissions.

Insight: Elderly patients appear to have a higher likelihood of repeated hospital utilization.

Recommendation: Strengthen discharge planning, follow-up care, and chronic disease management programs for elderly patients.

Average Length of Stay (LOS)

Result: 4.26 Days

Finding: Patients stayed an average of 4.26 days per admission.

Insight: Length of stay directly influences bed availability, staffing requirements, and treatment costs.

Recommendation: Monitor extended stays and identify opportunities to improve patient flow without compromising care quality.

Cost Analysis

Most Expensive Diagnosis Per Day

Diagnosis	Average Cost Per Day
Acute Appendicitis	47,500
Acute Myocardial Infarction	39,050
Chronic Kidney Disease	25,737.86

Finding: Acute Appendicitis generated the highest average treatment cost per day.

Insight: Patients with Acute Appendicitis required the greatest daily resource utilization compared to other diagnoses.

Recommendation: Review surgical workflows and post-operative resource allocation to ensure operational efficiency.

Cost Per Day Analysis

Formula:

Cost Per Day = Treatment Cost ÷ Length of Stay

Finding: Daily treatment costs varied significantly across admissions, ranging from approximately 7,333 to 48,750.

Insight: Cost Per Day provides a standardized measure of resource utilization regardless of patient length of stay.

Recommendation: Monitor patients and diagnoses with elevated daily costs to identify resource-intensive treatment patterns.

Key Findings

1. The hospital recorded 50 admissions from 18 unique patients.
 2. Essential Hypertension was the most common diagnosis with 9 recorded cases.
 3. Cardiology handled the highest patient volume with 14 admissions.
 4. Six elderly patients experienced repeated admissions, indicating a potentially high-risk population.
 5. Imran Ali generated the highest cumulative treatment cost at 835,000.
 6. Acute Appendicitis had the highest average daily treatment cost at 47,500.
 7. The average patient length of stay was 4.26 days.
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Recommendations

1. Strengthen hypertension prevention and management initiatives.
 2. Implement targeted follow-up programs for elderly patients with repeated admissions.
 3. Monitor high-cost patients and investigate treatment cost drivers.
 4. Ensure adequate staffing and resource allocation within Cardiology.
 5. Track Cost Per Day metrics to improve operational efficiency and resource utilization.
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Executive Summary

The hospital recorded 50 admissions involving 18 unique patients during the study period. Essential Hypertension was the most common diagnosis, while Cardiology managed the highest patient volume. Six elderly patients experienced repeated admissions, highlighting a potential opportunity to improve follow-up care and discharge planning. Acute Appendicitis generated the highest average treatment cost per day, indicating substantial resource utilization. The average patient length of stay was 4.26 days, and treatment costs remained a significant operational consideration. Management should prioritize reducing preventable readmissions, optimizing resource allocation, and monitoring high-cost diagnoses to improve both operational efficiency and patient outcomes.